

Practice Quiz — Strategic Thinking for Inventors

Part A: Multiple Choice

1. In Active Inference, a business model is best understood as:

- A) A fixed plan for generating revenue
- B) A generative model that predicts how value flows between your invention and its ecosystem
- C) A legal document describing corporate structure
- D) A marketing strategy for attracting customers

1. Blockbuster's failure to transition to streaming illustrates:

- A) Poor execution of a sound strategy
- B) The difficulty of model restructuring when the existing model generates accurate predictions for the current market
- C) Insufficient funding for technology investment
- D) Regulatory barriers to entering the streaming market

1. Scenario planning differs from single-point forecasting in that:

- A) It uses more data to make a more accurate single prediction
- B) It maintains multiple generative models of the future and selects strategies that perform well across several of them
- C) It is only useful for long-term planning, not short-term decisions
- D) It eliminates uncertainty about the future

1. When a technology approaches the top of its S-curve, an inventor should:

- A) Double down on optimizing the current technology
- B) Exit the market before the technology declines
- C) Begin investigating successor technologies while maintaining current competitiveness
- D) Lower prices to maintain market share

1. Strategic signaling involves:

- A) Sending coded messages to partners
- B) Taking observable actions designed to influence competitors' generative models of your strategy
- C) Publishing financial results to attract investors
- D) Broadcasting advertisements to potential customers

1. Jeff Bezos's "Day 1" philosophy is primarily a strategy for:

A) Maintaining early-stage pricing despite growth B) Keeping the company's generative model flexible and receptive to prediction errors C) Preserving the company's original business model indefinitely D) Hiring only employees who share the founder's vision

1. Confirmation bias in strategic thinking operates through:

A) Excessive precision on prior beliefs, causing genuine prediction errors to be dismissed rather than processed B) Lack of access to disconfirming information C) Deliberate deception by competitors D) Insufficient data collection

Part B: Short Answer and Analysis

1. Netflix evolved its business model from DVD-by-mail to streaming to original content production. For each transition, identify: (a) what prediction error triggered the model update, (b) what elements of the generative model changed, and (c) what new predictions the updated model generated.

2. SpaceX's approach to rocket reusability followed a sequence of increasingly ambitious tests (Grasshopper, sea landings, land landings). Explain this sequence as a cognitive strategy: why did SpaceX test incrementally rather than attempting a full reusable launch from the start? How does this approach relate to Active Inference principles of model updating under uncertainty?

3. You are the inventor of a novel water purification device for developing countries. Construct a 2x2 scenario matrix using two critical uncertainties: (a) whether international development funding increases or decreases, and (b) whether governments in target countries mandate water quality standards. For each of the four scenarios, describe the strategic implications and identify one robust strategy that works across all four.